LAKNIDU THILAKAWARDHANA

DEVOPS & SOFTWARE ENGINEER

laknidut@gmail.com | +94 768671725 | laknidu777.vercel.app | github.com/laknidu777 | linkedin.com/in/laknidu-thilakawardhana-411783247

PROFILE SUMMARY

DevOps and Software Engineer with proven expertise in orchestrating scalable cloud infrastructure, automated CI/CD pipelines, and high-performance full-stack applications. Experienced in Docker, Kubernetes, AWS, and modern microservices using .NET, NestJS, and Next.js. Dedicated to system reliability, infrastructure-as-code, and crafting frictionless developer workflows.

WORK EXPERIENCE

Intern Software and Devops Engineer - Koombiyo delivery **Jan 2024 - Present**

- Architected and managed scalable with Docker, cloud deployments on AWS with automated CI/CD pipelines using GitHub Actions, cutting release cycles.
- Implemented micro-frontend architectures with Next.js and optimized backend rendering pipelines, resulting in a 40% reduction in latency.

EDUCATION

BSc (Hons) in Computer Science (Second Upper Class Honours) **2022 - 2025**

NSBM - University of Plymouth

PROJECTS

Work Projects

COCO VANTAGE

NestJs, NextJs, Typescript, Docker, Prisma, PostgreSQL, Neon, Stripe

Developed an e-commerce platform for charcoal-based deodorizer products, enabling customers to browse products, place orders, make secure online payments through Stripe, and track their shipments. The platform includes an administrative dashboard for managing sales, inventory, and order fulfillment.

Implemented an online shopping and ordering experience to streamline product purchases and customer transactions.

Integrated the Stripe payment gateway to facilitate secure online payment processing.

Developed order and shipment tracking functionality to help customers monitor their purchases.

Built an administrative dashboard for centralized management of sales, inventory, and customer orders.

Streamlined e-commerce operations by bringing product management, order processing, and sales monitoring into one platform.

Key Areas: E-commerce Development, Stripe Payment Integration, Order Management, Shipment Tracking, Inventory Management, Sales Analytics, Admin Dashboard.

Kivi

Flutter, Dart, NestJs, PostgreSQL

Designed and developed a cloud-based Point-of-Sale (POS) system for small and medium-sized enterprises (SMEs), offering a subscription-based business model. Built to support sales and inventory operations across devices, the platform enables businesses to manage products, track inventory, and add items using mobile phones or PCs, with barcode scanning and both online and offline functionality.

Developed a cross-platform application using Flutter and backend services using NestJS, with PostgreSQL for data storage and management.

Implemented inventory management features to manage products, stock levels, and item information.

Designed role-based access management through an administrative panel, allowing business owners to grant appropriate access to multiple users.

Supported product entry and barcode scanning using mobile phones, alongside PC-based workflows.

Designed the system to support online and offline usage, enabling businesses to continue essential POS operations with limited or no internet connectivity.

Integrated GitHub Actions into the development workflow to automate CI/CD tasks and streamline software delivery.

Personal Projects

CloudOps Platform - DevOps Engineering Project (Ongoing)

Linux(Ubuntu/WSL), Docker, Docker compose, Git, Github, Github actions, GitHub Container Registry (GHCR), AWS Systems Manager, Node.js, AWS, EC2(

Developed and containerized a Node.js REST API with health-check endpoints and Docker health monitoring.

Implemented a CI pipeline using GitHub Actions to validate application files, run automated tests, build Docker images, and publish versioned images to GitHub Container Registry.

Deployed the containerized application to an AWS EC2 Ubuntu server and verified application and container health.

Configured AWS Systems Manager for remote instance management and command execution.

Practicing reliable deployment and rollback procedures, with automated deployment, security scanning, infrastructure as code, and monitoring planned as future enhancements.

Smart Home IoT Management System

TypeScript, NodeJs, NextJs, ExpressJS, PostgreSQL, C++, Arduino UNO, JavaScript, React Native

Designed and developed a smart home IoT management system for large buildings and organizations, enabling centralized monitoring and control of devices such as lights and fans. Built a web-based management platform using Next.js, Express.js, and React Native, integrated with ESP32 and Arduino UNO microcontrollers programmed in C++ for hardware-level device control.

Developed a centralized admin dashboard for managing devices, rooms, users, and access permissions across the building.

Implemented room management and user access control to ensure authorized individuals could manage designated spaces and devices.

Integrated IoT hardware with the software platform to enable remote control of connected electrical appliances.

Designed a scalable management solution to support multiple rooms, users, and connected devices within large building environments.

SKILLS

Cloud & DevOps: AWS, Docker, Kubernetes, Ansible, GitHub Actions, GitLab, Git, Nginx, Apache
Frontend & Mobile: Next.js, React Native, Flutter, JavaScript, Dart, Tailwind CSS,HTML, CSS
Backend & Services: .NET, NestJS, Node.js, C#, PHP, C++, REST APIs, ExpressJS,
Databases & ORM: PostgreSQL, MySQL, MongoDB, Microsoft SQL Server, Prisma, Firebase
Tools & Scripting: PowerShell, Postman, Vercel, Canva, Linux

REFERENCES

University of Plymouth Academic Reference

Department of Computing & Engineering
Phone: +44 (0)1752 585858
Email: computing@plymouth.ac.uk

Enterprise Systems Industry Reference

Technical Lead, Cloud & Infrastructure
Phone: +44 7700 900456
Email: director@enterprisetech.co.uk